# Prayas Sanyal

+91 87770 25465
prayassanyal008@gmail.com
Webpage | LinkedIn

## Education

2019–2023 **Heritage Institute of Technology, Kolkata**
B.Tech (Honours) Electronics and Communication Engineering
**GPA:** 9.44/10 [Rank: 5/211]
**Thesis:** Feature Detection for Onset of Alzheimer's Disease from Cognitively Normal Individuals

2019 **South Point High School, Kolkata**
AISSCE, Grade 12 (Percentage: **89.4%**) | AISSE, Grade 10 (Percentage: **93.1%**)

## Experience

### Research

2026 - Present **Indian Institute of Science, Bangalore**
Research Asssistant - **Advisor**: Dr. SP Arun
Project: Mechanisms of aesthetic perception of man-made shapes and natural scenes in humans and foundation models

- Conducted several psychophysics experiments with 100+ participants on their perception of beauty
- Fit behavioral data with pretrained foundation models to interpret aesthetic judgements
- Built computational models to understand compositional versus holistic representations underlying beauty perception in humans and deep neural networks

2022 - 2024 **Jadavpur University, Kolkata**
Research Associate - **Advisor**: Dr. Sanjoy Kumar Saha
Project: Semi-Supervised Disease Prediction from Retinal Fundus Images

- Curated a semi-supervised framework which locates disease manifestation in the retinal image
- Extracted disease relevant RoIs such as blood vessels, macula and optic cup and disc, red channel residual maps for downstream analysis
- F1 scores of 0.95 and 0.96 across two public datasets (JSIEC & RFMiD) covering diabetic retinopathy, hypertensive retinopathy, glaucoma, ARMD

2022 - 2023 **Heritage Institute of Technology, Kolkata**
Research Assistant - **Advisor**: Dr. Anindya Sen
Project: Temporal Study of Alzheimer's Disease Progression from Cognitively Normal

- Developed a wrapper pipeline in R to automatically extract tissue volumes - white matter, gray matter and cerebrospinal fluid for longitudinal analysis & perform temporal analysis on volume atrophy
- Standard pre-processing on sMRI; intensity normalization, bias field correction, skull stripping and image registration

### Industry

2024 - 2025 **Ernst & Young LLP**
Software Engineer

- C#, Angular and .NET developer: Responsible for developing features and maintaining full-stack web applications for automated financial management services

## Publications (* - equal contribution)

1 Prayas Sanyal, Srinjay Mukherjee, Arkapravo Das, Anindya Sen. "**Longitudinal Volumetric Study for the Progression of Alzheimer's Disease from Structural MRI**"
2024 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI), Prayagraj, India, 2024 (**Best Paper Award for Computer Vision**) Paper Code

2 Rohan Banerjee, Rakhshanda Mujib, Prayas Sanyal, Tapabrata Chakraborti, Sanjoy Kumar Saha." **Pan-Ret: A Semi-supervised Framework for Scalable Detection of Pan-Retinal Diseases**" Medical & Biological Engineering & Computing (2024). Paper

3 Rohan Banerjee, Rakhshanda Mujib, Prayas Sanyal, Sanjoy Kumar Saha. "**Detecting ARMD from Retinal Fundus Image: An RoI-Based Approach**"
International Conference on Artificial Intelligence & Sustainable Computing (AISC 2024) Paper

4 Ankan Basu, Jyotiraditya Roy*, Aditya Datta*, Prayas Sanyal*, Sumanta Banerjee. "**SolarTformer: A Transformer Based Deep Learning Approach for Short Term Solar Power Forecasting**" (Under Review ) Preprint

## Skills

**Languages** Python, C, C++, C#, LaTeX, R

**Libraries** NumPy, PyTorch, Tensorflow, FSL, Matplotlib, Seaborn; **Frameworks** Angular 20, .NET 8.0, Docker, jsPsych, Psychopy

**Relevant Courses** Digital Image Processing and Pattern Recognition, Data Structures, Object Oriented Programming, Intelligent Web and Big Data, Operating Sytems, Information Theory and Coding, DBMS, Advanced Numerical Methods, Signals and Systems, Digital Signal Processing, Microprocessors and Microcontrollers; **Audited:** Cognitive Neuroscience, Systems Neuroscience

## Academic and Extracurricular Activities

Qualified Graduate Aptitude Test in Engineering (GATE) CSE 2025, 2026

**TOEFL**: 6.0/6.0 (118/120)

**Certifications**

1. Microsoft Certified: Azure AI Fundamentals (Certification )
2. Mathematics for Machine Learning Specialization - Linear Algebra, Multivariate Calculus and Principal Component Analysis (Imperial College, Certification )
3. Neuroscience and Neuroimaging Specialization - Fundamental Neuroscience for Neuroimaging, Principles of fMRI 1, Principles of fMRI 2, Introduction to Neurohacking with R (John Hopkins University, Certification )

**Puzzle Solving** All India Rank 60 and State Rank 1 at Technothlon 2018 (IIT Guwahati), CS50x Puzzle Day by Harvard University, Massachusetts (2020 , 2023 , 2024 , 2025 )

## References

**Prof. SP Arun** | Professor, Centre for Neuroscience, Indian Institute of Science

**Prof. Sanjoy Kumar Saha** | Professor of Computer Science, Jadavpur University

**Prof. Anindya Sen** | Professor of Electronics and Communication, Heritage Institute of Technology